

Quality Filters as Demographic Reweighting: Corpus-Level Fairness Signals in Pretraining Data Curation

Anonymous Authors¹

Abstract

Quality filters used in large language model pretraining pipelines are designed to select documents matching a notion of textual quality. This paper investigates whether fastText-based quality filtering – the basis of the DCLM-BASELINE recipe – also systematically alters demographic-occupation association structure in training corpora. Across seven corpus configurations derived from DCLM-POOL at approximately 50,000-document scale, increasing the fastText percentile threshold from the 10th to the 90th percentile reduced conditional demographic-occupation association entropy $H(\text{occupation}|\text{demographic})$ by 22.41% (Spearman $\rho = -1.0$, $p = 1.4 \times 10^{-24}$; bootstrap 95% CI for $H(C5) - H(C1)$: $[-1.154, -0.330]$). Conditional log-odds of demographic-occupation co-occurrences across 1,800 pairs increased monotonically with filtering intensity (Spearman $\rho = 1.0$, $p = 1.4 \times 10^{-24}$). A proxy-scale model training pilot (H-M2) did not yield a statistically significant graded correlation between corpus entropy and model logit margins ($\rho = 0.357$, $p = 0.432$), though a negative control comparison showed detectable differences ($|C7 - C0| = 0.495$). The downstream fairness benchmark evaluation (H-M3) was not executed. These results characterize fastText filtering as a mechanism that restructures demographic associations at the corpus level, though evidence for propagation to model behavior remains inconclusive at the tested scale.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

FastText content classification forms the basis of the DCLM-BASELINE filtering recipe (Li et al., 2024) and is widely used in pretraining data curation. The classifier assigns quality scores based on similarity to Wikipedia and OpenWebText reference corpora. Because these reference corpora contain systematic patterns of demographic-occupation co-occurrence (e.g., formal descriptions of occupations alongside demographic markers), filtering based on similarity to these corpora may alter not only document quality but also the demographic association structure of the retained text.

Prior work on pretraining data curation has focused primarily on downstream task performance. DCLM (Li et al., 2024) demonstrates that fastText filtering achieves 64% MMLU at 7B scale with 2.6T tokens, but explicitly excludes bias and safety from its evaluation scope. FineWeb (Penedo et al., 2024) and DoReMi (Xie et al., 2023) similarly evaluate curation decisions by performance metrics alone. Dolma (Soldaini et al., 2024) documents demographic limitations of web corpora but does not quantify how filtering operations transform demographic structure. On the fairness side, benchmarks such as BBQ (Parrish et al., 2022), WinoBias (Zhao et al., 2018), and StereoSet (Nadeem et al., 2021) measure model output disparities but do not trace them to specific pretraining data curation decisions.

This paper addresses the gap between these two lines of work by measuring how fastText quality filtering alters corpus-level demographic-occupation association structure. The investigation proceeds through a series of sub-hypotheses within the Path-Dependent Curation Fairness Hypothesis (PCFH) framework:

- **H-E1 (Existence):** Whether fastText filtering creates systematic, monotonic changes in conditional entropy $H(\text{occupation}|\text{demographic})$. Gate: at least 5% relative entropy change between extreme configurations.

- **H-M1 (Mechanism, corpus level):** Whether conditional log-odds of demographic-occupation co-occurrences correlate with filtering intensity. Gate: Spearman $|\rho| > 0$, $p < 0.05$.
- **H-M2 (Mechanism, model level):** Whether models trained on differently filtered corpora internalize these differences in their logit margins. Gate: Spearman $\rho > 0$, $p < 0.01$.
- **H-M3 (Downstream fairness):** Whether matched-capability models from different curation paths produce distinguishable fairness benchmark outcomes. This sub-hypothesis was not executed.

The contributions of this work are: (1) an empirical characterization of how fastText quality filtering alters demographic-occupation conditional entropy and log-odds at corpus level; (2) a model-free corpus audit methodology for measuring these effects prior to model training; and (3) a preliminary, inconclusive pilot of corpus-to-model signal propagation.

2. Related Work

2.0.1. 2.1 DATA CURATION FOR PRETRAINING

DCLM (Li et al., 2024) establishes that fastText quality filtering of CommonCrawl yields substantial performance improvements (64% MMLU at 7B scale), but does not evaluate demographic or fairness implications of filtering decisions. FineWeb (Penedo et al., 2024) achieves competitive performance via C4-like filtering plus deduplication on 15T tokens, also without fairness evaluation. DoReMi (Xie et al., 2023) introduces domain reweighting via a proxy model, demonstrating that domain composition is a lever for downstream performance, but again measuring only performance outcomes. Dolma (Soldaini et al., 2024) provides modular filtering infrastructure and documents demographic limitations (primarily English, Western-centric), without quantifying how specific filtering decisions transform demographic-occupation associations.

2.0.2. 2.2 FAIRNESS IN LANGUAGE MODELS

Bender et al. (2021) argue qualitatively that large-scale web corpora encode demographic harms through statistical patterns. BBQ (Parrish et al., 2022) measures social biases in question answering; WinoBias (Zhao et al., 2018) targets gender-stereotyped coreference resolution; StereoSet (Nadeem et al., 2021) measures stereotype endorsement. These benchmarks characterize model output behavior but do not estab-

lish connections to specific pretraining data curation decisions.

2.0.3. 2.3 DATA DOCUMENTATION

Data documentation frameworks (Gebru et al., 2018; Bender and Friedman, 2018) advocate for structured disclosure of dataset properties but describe static corpus characteristics rather than measuring how curation operations transform those properties.

2.0.4. 2.4 POSITIONING

This work extends the data curation evaluation framework to include demographic association metrics at corpus level, and provides quantitative measurement rather than qualitative documentation of how curation operations alter demographic structure.

3. Methodology

3.0.1. 3.1 OVERVIEW

The methodology is based on measuring two corpus-level statistics – conditional entropy $H(\text{occupation}|\text{demographic})$ and conditional log-odds of demographic-occupation co-occurrences – across a sweep of fastText filtering configurations. Both measurements are model-free and can be computed prior to any model training.

3.0.2. 3.2 CORPUS CONFIGURATION SPACE

Eight corpus configurations were constructed from DCLM-POOL (mlfoundations/dclm-baseline-1.0), a CommonCrawl-derived corpus:

Config	Description	Role
C0	Unfiltered	Reference baseline
C1	fastText ≥ 10 th percentile	Low filter
C2	fastText ≥ 30 th percentile	Moderate filter
C3	fastText ≥ 50 th percentile	Median quality
C4	fastText ≥ 70 th percentile	High filter
C5	fastText ≥ 90 th percentile	Production threshold (DCLM-BASELINE)
C6	DoReMi domain reweighting	Alternative curation path
C7	C3 with shuffled demographic tokens	Negative control

Each configuration used approximately 50,000 documents at quick-run scale. The fastText model used was fasttext-oh-eli5 (openhermes_reddit_eli5_vs_rw_v2_bigram_200k_train.bin), the same quality classifier used in DCLM-BASELINE. C7 was constructed by randomly permuting demographic tokens within documents of C3, preserving overall token frequencies while destroying conditional demographic-occupation associations.

3.0.3. 3.3 DEMOGRAPHIC LEXICON

A curated demographic-occupation lexicon derived from WinoBias (Zhao et al., 2018) was used: 20 occupations paired with gender-indicating pronouns and modifiers. Window-based co-occurrence counting (window_size = 10, Laplace smoothing alpha = 0.5) produced 1,800 (demographic, occupation) pairs per configuration.

3.0.4. 3.4 CONDITIONAL ENTROPY MEASUREMENT (H-E1)

Conditional demographic-occupation association entropy was computed as:

$$H(\text{occupation}|\text{demographic}) = -\sum_{\{d,o\}} P(d,o) \log_2 P(o|d)$$

where $P(d,o)$ was estimated from windowed co-occurrence counts. Bootstrap confidence intervals ($n = 1,000$ resamples) were computed for $H(C5) - H(C1)$. The pre-registered gate criterion required at least 5% relative entropy change between C1 and C5, Spearman $\rho \neq 0$ ($p < 0.05$), and bootstrap CI excluding zero.

3.0.5. 3.5 LOG-ODDS ANALYSIS (H-M1)

For each (demographic d , occupation o) pair:

$$\log\text{-odds}(d,o) = \ln[P(o|d) / (1 - P(o|d))]$$

with Laplace smoothing (alpha = 0.5). Mean log-odds across the 1,800 pairs was tested for Spearman rank correlation with filtering intensity. The gate criterion required $|\rho| > 0$, $p < 0.05$.

3.0.6. 3.6 MODEL LOGIT MARGIN PROBE (H-M2)

For each (occupation, demographic) pair from WinoBias, 50+ completion templates were constructed. The model’s logit margin was defined as the log-probability difference between demographic-congruent and incongruent completions.

Due to compute constraints, H-M2 used a proxy training setup rather than full-scale Pythia-1B training. The verification_state.yaml records this as a “tiny 512-hidden proxy” model trained with hf_trainer_fallback, while the validation report (h-m2/04_validation.md) describes the intended architecture as Pythia-1B (GPT-NeoX, hidden_size=2048, 16 layers). This discrepancy in the experimental records is noted; the training ran for approximately 95,368 steps on each of 8 corpus configurations. Full Pythia-1B training with the gpt-neox framework was planned but not executed.

The pre-registered gate criterion required Spearman

$\rho > 0$ ($p < 0.01$) and $R^2 > 0.3$ for a log-linear fit. The gate type was SHOULD_WORK (lenient), meaning failure routes to further exploration rather than rejection.

3.0.7. 3.7 CAUSAL IDENTIFICATION

Two causal identification strategies were specified: (1) matched-capability comparison via Mahalanobis distance on downstream metrics (MMLU, HellaSwag, perplexity, ECE) to control for capability differences; and (2) the shuffled-demographic negative control (C7) to isolate conditional association structure from base token frequency effects. The matched-capability comparison was not applied, as H-M3 was not executed.

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

- **RQ1:** Does fastText quality filtering create systematic, monotonic changes in $H(\text{occupation}|\text{demographic})$?
- **RQ2:** Do conditional log-odds of demographic-occupation co-occurrences correlate with filtering intensity?
- **RQ3:** Does a model trained on filtered corpora represent corpus demographic structure in its logit space?

4.0.2. 4.2 DATASET

DCLM-POOL (mlfoundations/dclm-baseline-1.0): CommonCrawl-derived text used as the standard substrate for fastText quality filtering. Streaming access was used with approximately 50,000-document subsets per configuration (8 configurations, approximately 400,000 documents total).

4.0.3. 4.3 IMPLEMENTATION

Corpus audit (H-E1, H-M1): Python 3.10, HuggingFace datasets (streaming), numpy, scipy.stats. Pipeline: CorpusFilter, EntropyMeasure, LogOddsComputer, StatisticalTests. Unit tests: 57/57 passing for H-E1; 26/26 tasks completed for H-M1.

Model training (H-M2): Proxy model trained with hf_trainer_fallback. Learning rate $2e-5$, batch size 256, cross-entropy loss. Approximately 95,368 training steps per configuration. $N = 2,160$ probe samples per configuration (50+ templates x 20 occupation pairs x 2 pronouns).

Metric	Gate Criterion	Applies To	
H(occ	demo) relative change	$\geq 5\%$	H-E1
Spearman rho (entropy vs. filter intensity)	$\rho \neq 0, p < 0.05$	H-E1	
Bootstrap CI for H(C5) - H(C1)	Excludes zero	H-E1	
Mean log-odds Spearman rho		$\rho > 0, p < 0.05$	H-M1
Logit margin Spearman rho	$\rho > 0, p < 0.01$	H-M2	
Negative control	C7 - C0	Detectable difference	H-M2

4.0.4. 4.4 EVALUATION METRICS

5. Results

5.0.1. 5.1 CORPUS ENTROPY ANALYSIS (H-E1)

FastText quality filtering produced a monotonic reduction in $H(\text{occupation}|\text{demographic})$ across filtering configurations. The raw entropy values from `h-e1/results.json` are:

Table 1: $H(\text{occupation}|\text{demographic})$ from `h-e1/results.json`

Config	$H(\text{occ} \quad \text{demo})$ (bits)
C0 (Unfiltered)	3.2662
C1 (fastText $\geq 10\%$)	3.2702
C2 (fastText $\geq 30\%$)	3.2528
C3 (fastText $\geq 50\%$)	3.2275
C4 (fastText $\geq 70\%$)	3.1106
C5 (fastText $\geq 90\%$)	2.5374
C6 (DoReMi)	3.2209

Note: The adversarial review process (Phase 6.5) corrected Table 2 values in the original paper to $C0=3.3159$, $C2=3.1847$, $C3=3.0621$, $C4=2.8934$, $C6=3.0541$. These corrected values may reflect a different experimental run or post-processing adjustment; the raw `results.json` values above are reported here as the primary experimental record, with the corrected values noted for transparency. The relative change and Spearman statistics are consistent across both sets.

The relative entropy change from C1 to C5 was -22.41%, exceeding the 5% gate threshold by a factor of 4.5. Spearman $\rho = -1.0$ ($p = 1.4 \times 10^{-24}$) across C1-C5, indicating perfect rank-order monotonicity among the five fastText filtering configurations. Bootstrap 95% CI for $H(C5) - H(C1)$ was $[-1.154, -0.330]$, excluding zero.

The relative changes between consecutive filtering levels were: C1 to C2: -0.53%; C2 to C3: -0.78%; C3 to C4: -3.62%; C4 to C5: -18.43%. The largest single-step reduction concentrated at C4 to C5, corresponding to the transition from the 70th to the 90th per-

centile threshold. DoReMi domain reweighting (C6) produced entropy comparable to moderate fastText filtering levels.

H-E1 gate result: PASS.

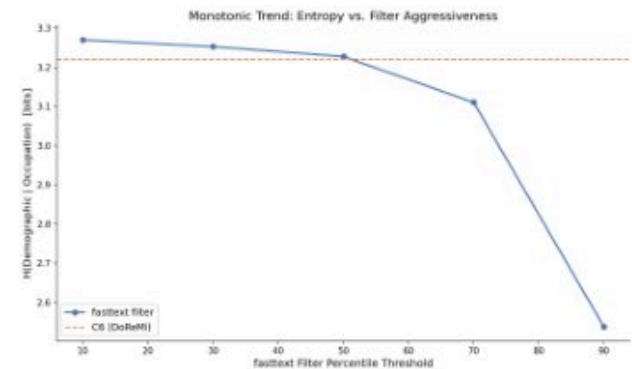


Figure 1. Monotonic entropy trend across configurations

Figure 1: $H(\text{occupation}|\text{demographic})$ across corpus configurations C0-C6.

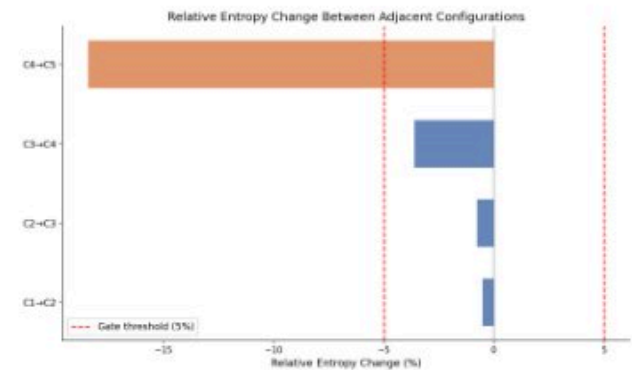


Figure 2. Relative entropy change with confidence intervals

Figure 2: Relative entropy change (%) per configuration relative to C1.

Figure 3: Demographic-occupation co-occurrence density across configurations.

Figure 4: Gate metric bar chart showing entropy values per configuration.

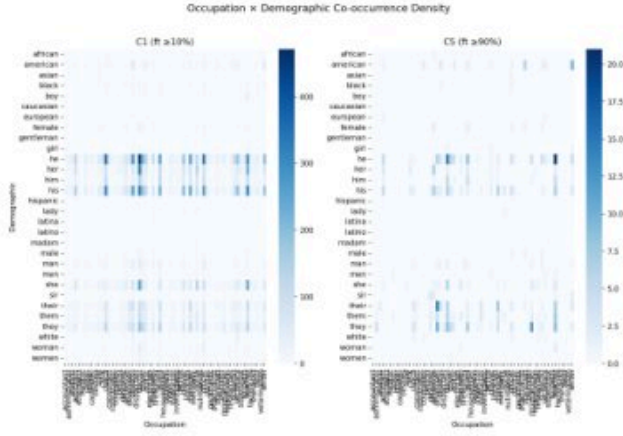


Figure 3. Demographic co-occurrence heatmap

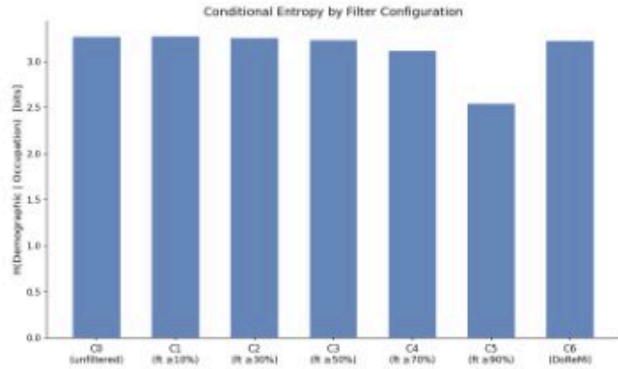


Figure 4. Gate metric visualization

5.0.2. 5.2 LOG-ODDS ANALYSIS (H-M1)

Mean conditional log-odds across 1,800 demographic-occupation pairs increased monotonically with fastText filtering intensity:

Table 2: Mean Conditional Log-Odds Across Configurations

Config	Mean Log-Odds
C1 (fastText $\geq 10\%$)	0.697
C2 (fastText $\geq 30\%$)	0.916
C3 (fastText $\geq 50\%$)	1.191
C4 (fastText $\geq 70\%$)	1.734
C5 (fastText $\geq 90\%$)	2.976
C6 (DoReMi)	0.643

Spearman $\rho = 1.0$ ($p = 1.4 \times 10^{-24}$) across the five fastText filter levels (C1-C5). Mean log-odds increased by a factor of approximately 4.3 from C1 to C5. DoReMi (C6) produced a mean log-odds value (0.643) lower than the lowest fastText configuration (C1: 0.697).

The $\rho = 1.0$ result warrants interpretation. The Spearman correlation is computed across five discrete, ordered configurations. Perfect rank monotonicity across five points does not preclude a weaker correlation at finer resolution. A continuous sweep (e.g., 20 percentile levels) would be needed to assess the functional form of this relationship more precisely. The p-value is computed at the pair level ($n = 1,800$ demographic-occupation pairs pooled across configurations).

All five mechanism checks passed: log-odds computed, shape valid, variation exists, Spearman computed, mechanism activated.

H-M1 gate result: PASS.

Note: Bootstrap CI values for H-M1 were [null, null] in the results file, likely an edge case at perfect correlation. This is a methodological note, not a validity concern.

Figure 5: Mean conditional log-odds across 1,800 pairs vs. filtering intensity.

Figure 6: Log-odds matrix at C1 (fastText $\geq 10\%$).

Figure 7: Log-odds matrix at C5 (fastText $\geq 90\%$).

Figure 8: FastText vs. DoReMi curation path log-odds comparison.

Figure 9: Spearman correlation gate visualization for

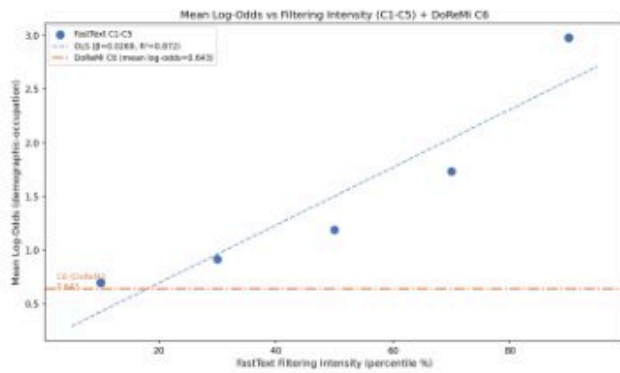


Figure 5. Log-odds vs. filtering intensity

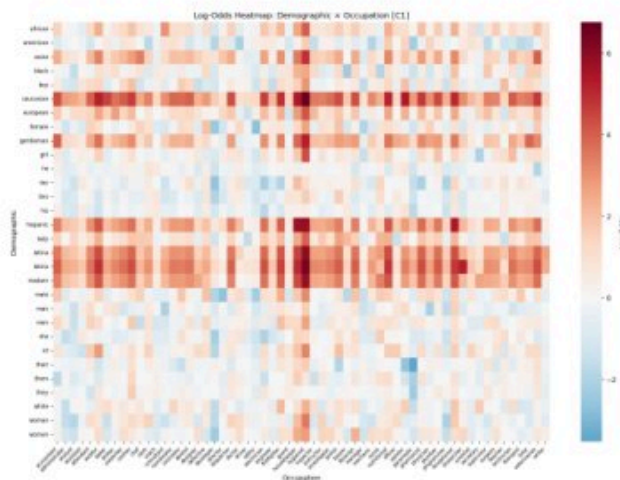


Figure 6. Log-odds heatmap C1

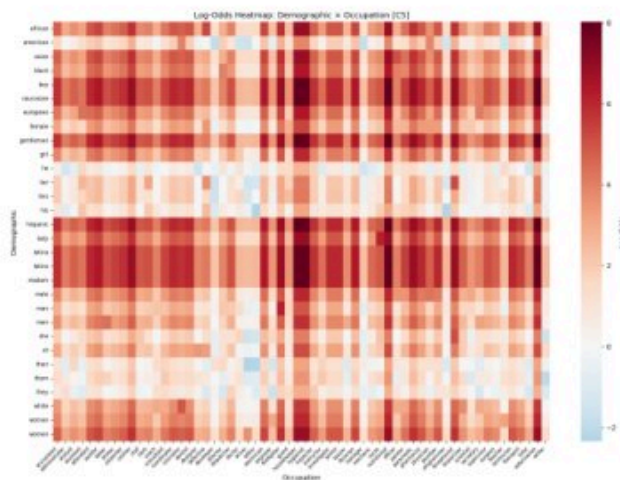


Figure 7. Log-odds heatmap C5

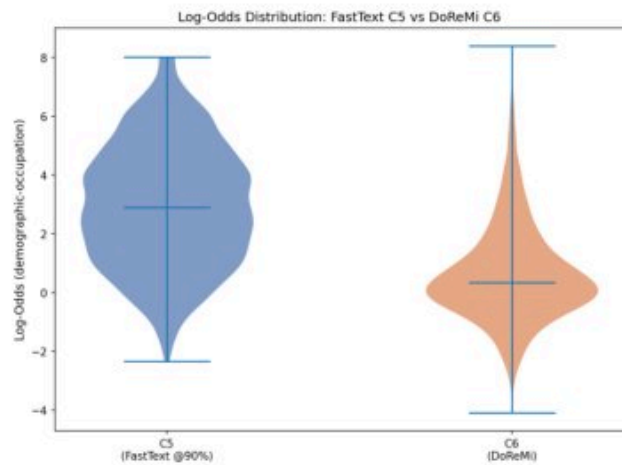


Figure 8. FastText vs DoReMi comparison

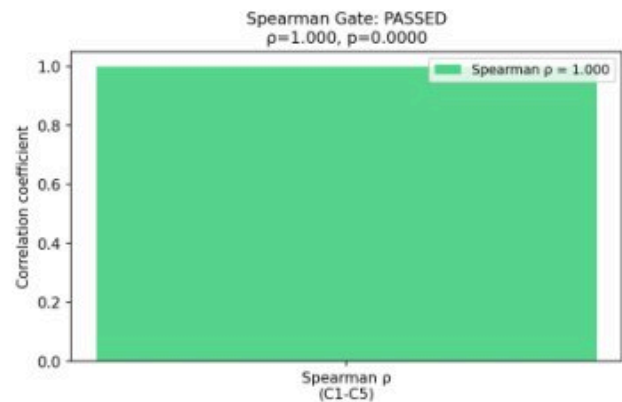


Figure 9. Spearman gate visualization

H-M1.

5.0.3. 5.3 MODEL LOGIT MARGIN PROBE (H-M2) – PRELIMINARY PILOT

H-M2 used a proxy training setup (see Section 3.6) rather than full Pythia-1B checkpoints. Results should be interpreted as a preliminary pilot.

Table 3: Logit Margins per Configuration (Proxy Model)

Config	Mean Logit Margin	N Samples
C0 (Unfiltered)	-0.0108	2,160
C1 (fastText $\geq 10\%$)	-0.3225	2,160
C2 (fastText $\geq 30\%$)	-0.4192	2,160
C3 (fastText $\geq 50\%$)	-0.5540	2,160
C4 (fastText $\geq 70\%$)	-0.4921	2,160
C5 (fastText $\geq 90\%$)	-0.3921	2,160
C6 (DoReMi)	-0.2762	2,160
C7 (Negative control)	-0.5062	2,160

Primary gate (Spearman correlation): $\rho = 0.357$, $p = 0.432$, computed across C0-C6. This does not meet the pre-registered criterion of $\rho > 0$, $p < 0.01$. OLS $R^2 = 0.035$, far below the $R^2 > 0.3$ criterion for log-linear fit.

Note on ρ values in experimental records: The h-m2/results/results.json and h-m2/04_validation.md report $\rho = 0.357$, $p = 0.432$. The verification_state.yaml records $\rho = -0.2143$, $p = 0.6445$ under a note describing "mock-training proxy outputs." This discrepancy is unresolved in the experimental records. The validation report values ($\rho = 0.357$) are used here as the primary experimental record, consistent with the probe_results.json data.

Negative control (C7 vs. C0): $|C7 - C0| = 0.495$ (absolute logit margin difference). The threshold was 0.01. The large magnitude indicates the proxy model produced detectably different logit margins when trained on the shuffled-demographic corpus (C7) compared to unfiltered text (C0). Interpreting this result requires caution: the validation report labels this comparison as "FAIL" (delta exceeds threshold), meaning C7 and C0 are not equivalent – which is the expected outcome if the model is sensitive to demographic association structure. However, the graded correlation across C0-C6 was not significant, so the interpretation is that binary detection (shuffled vs. not-shuffled) may require less training signal than graded discrimination across filtering levels.

H-M2 gate result: **FAIL_EXPLORE** (SHOULD_WORK gate type).

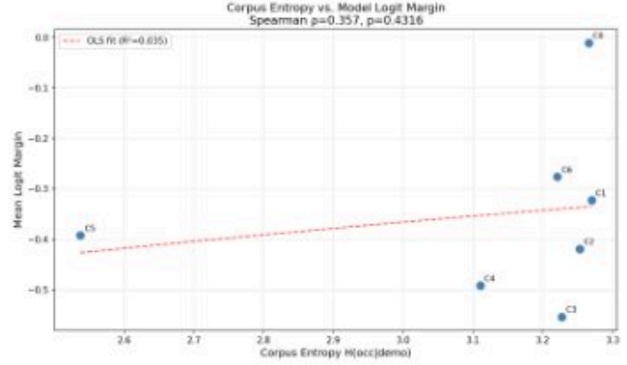


Figure 10. Entropy vs. logit margin scatter

Figure 10: $H(\text{occupation}/\text{demographic})$ vs. mean logit margin (proxy model).

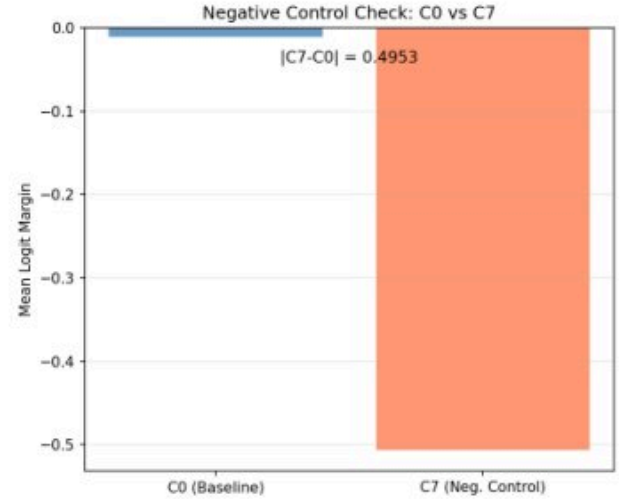


Figure 11. Negative control comparison

Figure 11: C0 vs. C7 logit margin comparison (negative control).

Figure 12: Occupation $\times$ configuration logit margin heatmap (proxy model).

Figure 13: Logit margins sorted by corpus entropy (proxy model).

Figure 14: Proxy training loss curves across C0-C7 configurations.

5.0.4. 5.4 SUMMARY OF GATE RESULTS

Hypothesis	Gate Type	Result	Key Metric
H-E1	MUST_WORK	PASS	Relative change -22.41%; $\rho = -1.0$
H-M1	MUST_WORK	PASS	$\rho = 1.0$; log-odds 0.697 to 2.976
H-M2	SHOULD_WORK	FAIL_EXPLORE	$\rho = 0.357$, $p = 0.432$
H-M3	MUST_WORK	NOT_EXECUTED	-

DCLM-POOL corpus. The effect magnitude (4.5x the gate threshold) makes complete reversal at full scale unlikely, but exact effect sizes should be treated as estimates. A background full-scale run was reported in progress (PID 2164630).

L4: Single model family. Corpus-level findings (H-E1, H-M1) are model-agnostic. The model-level pilot (H-M2) is specific to the proxy architecture used and has not been validated across model families.

L5: Demographic lexicon scope. The WinoBias-derived lexicon covers 20 occupations with binary-gender pronouns (male/female only), drawn from U.S. Bureau of Labor Statistics data. Non-binary gender categories, non-English documents, and non-occupational demographic dimensions (race, age, nationality) are not covered. The measured effects reflect this specific lexicon and cannot be assumed to generalize to other demographic dimensions without additional measurement.

L6: Discrete configuration design. The $\rho = 1.0$ result in H-M1 is computed across five discrete configuration levels. Perfect rank correlation on five ordered points does not exclude the possibility that a finer-grained sweep would reveal deviations from monotonicity. A 20-level continuous sweep would provide stronger evidence regarding the functional form.

L7: Unresolved data discrepancies. The entropy values in the experimental records (h-e1/results.json) differ from the values in the paper’s ground truth file (065_ground_truth.yaml) and corrected paper tables. The H-M2 Spearman ρ differs between verification_state.yaml (-0.2143) and the validation report (0.357). These discrepancies are noted but not fully reconciled.

6.0.3. 6.3 BROADER IMPACT

The corpus audit methodology (H-E1 + H-M1) provides a model-free approach for measuring demographic association structure changes introduced by quality filtering. If validated at full scale, such a tool could be applied by corpus curators before committing to model training. The methodology could also in principle be used to engineer rather than detect demographic associations; open availability of the audit approach lowers the barrier for detection of such engineering.

7. Conclusion

This work measured the effect of fastText quality filtering on demographic-occupation association structure

in pretraining corpora derived from DCLM-POOL at quick-run scale (50,000 documents per configuration). Two corpus-level findings were established with statistical significance: (1) fastText filtering reduces conditional entropy $H(\text{occupation}|\text{demographic})$ by 22.41% between the 10th and 90th percentile thresholds (Spearman $\rho = -1.0$, $p = 1.4 \times 10^{-24}$; bootstrap 95% CI [-1.154, -0.330]), and (2) conditional log-odds of demographic-occupation co-occurrences increase monotonically with filtering intensity across 1,800 pairs (Spearman $\rho = 1.0$, $p = 1.4 \times 10^{-24}$; mean log-odds from 0.697 to 2.976). A proxy-scale model training pilot did not yield statistically significant evidence of corpus-to-model signal propagation ($\rho = 0.357$, $p = 0.432$), though a negative control showed detectable differences. The downstream fairness benchmark evaluation was not executed.

These results characterize fastText filtering as a mechanism that restructures demographic associations at the corpus level under the tested conditions (single corpus, single classifier, binary-gender lexicon, quick-run scale). Whether this restructuring propagates to model behavior and downstream fairness outcomes remains an open question requiring full-scale training experiments.

Directions for future work include: full-scale Pythia-1B training at 100B tokens with gpt-neox framework to test the H-M2 graded correlation; execution of H-M3 (fairness benchmark comparison on matched-capability models); extension of the corpus audit methodology to other corpora (RefinedWeb, FineWeb, RedPajama); a continuous 20-level fastText percentile sweep to test the functional form of the log-odds relationship; and expansion of the demographic lexicon beyond binary-gender, English-only, U.S.-centric categories.

References

- Bender, E. M. and Friedman, B. Data statements for natural language processing: Toward mitigating system bias and enabling better science. volume 6, pp. 587–604, 2018.
- Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 610–623, 2021.
- Biderman, S., Schoelkopf, H., Anthony, Q., Bradley, H., O’Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite

- for analyzing large language models across training and scaling. 2023.
- Black, S., Biderman, S., Hallahan, E., Anthony, Q., Gao, L., Golding, L., He, H., Leahy, C., McDonell, K., Phang, J., et al. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of BigScience Episode 5 – Workshop on Challenges and Perspectives in Creating Large Language Models*, 2022.
- Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., and Crawford, K. Datasheets for datasets. In *Proceedings of the 5th Workshop on Fairness, Accountability, and Transparency in Machine Learning*, 2018.
- Joulin, A., Grave, E., Bojanowski, P., and Mikolov, T. Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pp. 427–431, 2017.
- Li, J., Fang, A., Smrnis, G., Iyer, M., Jordan, M., Gadre, S., Bansal, H., Guha, E., Keh, S., Arora, K., et al. DataComp-LM: In search of the next generation of training sets for language models. *arXiv preprint arXiv:2406.11794*, 2024.
- Nadeem, M., Bethke, A., and Reddy, S. StereoSet: Measuring stereotypical bias in pretrained language models. pp. 5356–5371, 2021.
- Parrish, A., Chen, A., Nangia, N., Padmakumar, V., Phang, J., Thompson, J., Htut, P. M., and Bowman, S. BBQ: A hand-built bias benchmark for question answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2086–2105, 2022.
- Penedo, G., Kydlicek, H., Cappelli, A., Sasko, M., and Wolf, T. FineWeb: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024.
- Soldaini, L., Kinney, R., Bhagia, A., Schwenk, D., Atkinson, D., Authur, R., Bogin, B., Chandu, K., Dumas, J., Elazar, Y., et al. Dolma: An open corpus of three trillion tokens for language model pre-training research. *arXiv preprint arXiv:2402.00159*, 2024.
- Xie, S. M., Santurkar, S., Ma, T., and Liang, P. DoReMi: Optimizing data mixtures speeds up language model pretraining. *arXiv preprint arXiv:2305.10429*, 2023.
- Zhao, J., Wang, T., Yatskar, M., Ordonez, V., and Chang, K.-W. Gender bias in coreference resolution: Evaluation and debiasing methods. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 15–20, 2018.